



PRESIDENCY UNIVERSITY

BENGALURU

EVEN SEMESTER – (2025-2026), SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

QUESTION BANK- ENDTERM

Course Title : Machine Vision for Robotics Course code : ECE3107
Instructor In-charge : Dr. K. Gowri Semester : VII
Date : Max. Marks : 30

The conduction of Endterm will be in offline mode in the University Campus.

QUESTION BANK

Part A

S.No.	Questions	CO
1	What is Machine Vision?	CO1
2	Define feature of an image.	CO1
3	What is contrast enhancement?	CO2
4	Write any two examples of features in any image.	CO3
5	What is template matching?	CO3
6	Why YCrCb components are preferred in image analysis of natural images rather than RGB components?	CO4
7	Write any two algorithms used for image classification tasks.	CO4
8	Write any two tracking algorithms.	CO5
9	List the three ways to simulate a robot in ROS.	CO6
10	What are the three pillars of open CV, open NI and PCL?	CO6
11	Define an image.	CO1
12	Mention the different types of gray value transformations.	CO1
13	Why image smoothing is important in image processing?	CO2
14	What is stereo reconstruction?	CO3
15	What is similarity transformation?	CO3
16	What is transfer learning?	CO4
17	What is iconic image processing?	CO4
18	Define histogram of an image.	CO5
19	What are the advantages of LED diodes?	CO1
20	What is the focal length of a camera?	CO1
21	Define dispersion.	CO2
22	What is rigid transformation?	CO3

23	Write any two challenges in object recognition using a single view.	CO3
24	What is multi image processing?	CO4
25	What are SIFT features?	CO4
26	What is bin number in spatiograms?	CO4

Part B

S.No.	Questions	CO
1	Write about the components of typical machine vision system	CO1
2	Explain in detail about different types of light sources used in machine vision	CO1
3	Explain how an object can be segmented using thresholding method.	CO2
4	Summarize the method of gray value transformation and contrast enhancement in image enhancement.	CO2
5	Write about any one of the digital video connectors of computer camera interface.	CO2
6	Why camera calibration is important? Explain how camera calibration is done.	CO2
7	Describe in detail about system component of an object recognition.	CO3
8	Explain stereo reconstruction.	CO3
9	Explain the recognition by linear combination of models using two views only.	CO3
10	Explain the steps of an object recognition using only a single view.	CO3
11	List the steps of transforming sensors reading into values.	CO4
12	Describe about mapping SONAR data.	CO4
13	Explain how the LASER scan measurements are aligned.	CO4
14	What are the two issues that make computer vision difficult? Explain.	CO4
15	Why classification algorithms are used in landmark spatiograms?	CO5
16	Enumerate the steps of condensation algorithm.	CO5
17	Explain in detail about spatial histograms. How it differs from histogram of an image?	CO5
18	Explain the unsupervised learning algorithm: K-means clustering.	CO5
19	Write in detail about installing and testing of ROS camera drivers.	CO6
20	Explain cv_bridge package.	CO6
21	Write about the elements of visual perception.	CO1
22	Discuss about contrast normalization and robust contrast normalization, a method of image enhancement.	CO2
23	Explain in detail about different types of camera sensors.	CO1
24	Describe segmentation of lines, circles and contours.	CO2
25	Explain in detail about pin hole cameras.	CO1

26	Explain how to correct the position, rotation, scaling, skew and translations of an image.	CO2
27	Describe the fundamental data structures: image, region, subpixel precise contours	CO1
28	What is image smoothing? Explain.	CO2



Signature of Instructor in charge